

KKT Conditions and Lagrangian Duality

Complete Two-Lecture Notes on Constrained and Convex Optimization

Contents

How to Use These Notes

These notes are designed for approximately two lectures of a course in optimization, particularly for students encountering constrained optimization, KKT conditions, convex optimization, and duality for the first time.

The material is organized as follows:

- **Lecture 1:** From unconstrained optimization to Lagrange multipliers and KKT conditions.
- **Lecture 2:** Lagrangian duality, weak and strong duality, Slater's condition, and the relationship between KKT and optimality.

Throughout the notes, inequality constraints are written in the standard form

$$g_i(x) \leq 0.$$

This sign convention is important because it determines the sign of the Lagrange multipliers.

1 Lecture 1: KKT Conditions

1.1 Motivation: Why Do We Need KKT Conditions?

Consider first the unconstrained optimization problem

$$\min_{x \in \mathbb{R}^n} f(x).$$

If f is differentiable and x^* is an interior local minimizer, then the first-order necessary condition is

$$\boxed{\nabla f(x^*) = 0.}$$

This is the familiar condition from elementary calculus.

However, suppose that we impose constraints:

$$\begin{aligned} \min_x \quad & f(x) \\ \text{subject to} \quad & g_i(x) \leq 0, \quad i = 1, \dots, m, \\ & h_j(x) = 0, \quad j = 1, \dots, p. \end{aligned}$$

Now the optimum may occur on the boundary of the feasible region.

At a boundary point, $\nabla f(x^*)$ does not necessarily have to be zero.

Intuition

At an unconstrained minimum, we are free to move in every sufficiently small direction. Therefore there can be no direction of decrease, giving $\nabla f(x^*) = 0$.

At a constrained minimum, some directions may be forbidden by the constraints. Therefore the gradient of the objective need not vanish. Instead, the objective gradient must be balanced by the normals to the active constraints.

This is the fundamental idea behind the KKT conditions.

1.2 Unconstrained Optimization

Consider

$$\min_x f(x).$$

Suppose f is differentiable and x^* is an interior local minimizer.

For any direction d , define

$$\phi(t) = f(x^* + td).$$

Since x^* is a local minimum,

$$\phi'(0) = 0.$$

By the chain rule,

$$\phi'(0) = \nabla f(x^*)^\top d.$$

Since this must hold for every $d \in \mathbb{R}^n$,

$$\nabla f(x^*) = 0.$$

Thus

$$\boxed{\nabla f(x^*) = 0}$$

is the first-order necessary condition for an unconstrained differentiable optimization problem.

1.3 Equality-Constrained Optimization

Consider

$$\begin{aligned} \min_x \quad & f(x) \\ \text{subject to} \quad & h(x) = 0. \end{aligned}$$

Suppose x^* is a local constrained optimum.

The constraint defines a surface

$$h(x) = 0.$$

The vector

$$\nabla h(x^*)$$

is normal to this surface.

A feasible direction d must be tangent to the constraint surface. Hence, to first order,

$$\nabla h(x^*)^\top d = 0.$$

At the optimum, the directional derivative of f must vanish in every feasible tangent direction:

$$\nabla f(x^*)^\top d = 0.$$

Thus $\nabla f(x^*)$ and $\nabla h(x^*)$ belong to the same normal space. Under a regularity condition,

$$\nabla f(x^*) = \lambda \nabla h(x^*).$$

Equivalently,

$$\boxed{\nabla f(x^*) - \lambda \nabla h(x^*) = 0.}$$

This is the method of Lagrange multipliers.

1.4 The Lagrangian

Define

$$\boxed{\mathcal{L}(x, \lambda) = f(x) + \lambda h(x).}$$

Then the first-order conditions are

$$\nabla_x \mathcal{L}(x^*, \lambda^*) = 0$$

together with

$$h(x^*) = 0.$$

For several equality constraints,

$$\begin{aligned} \min_x \quad & f(x) \\ \text{subject to} \quad & h_j(x) = 0, \quad j = 1, \dots, p, \end{aligned}$$

the Lagrangian becomes

$$\boxed{\mathcal{L}(x, \mu) = f(x) + \sum_{j=1}^p \mu_j h_j(x).}$$

The stationarity condition is

$$\boxed{\nabla f(x^*) + \sum_{j=1}^p \mu_j^* \nabla h_j(x^*) = 0.}$$

The equality multipliers μ_j can have either sign.

1.5 Why Inequality Constraints Are Different

Now consider

$$\min_x f(x)$$

subject to

$$g(x) \leq 0.$$

An inequality constraint divides space into a feasible side and an infeasible side.

There are two possibilities at a feasible point x^* .

1.5.1 Active Constraint

The constraint is active if

$$g(x^*) = 0.$$

The point lies exactly on the boundary.

1.5.2 Inactive Constraint

The constraint is inactive if

$$g(x^*) < 0.$$

The point lies strictly inside the feasible side.

Intuition

An inactive constraint does not locally restrict the optimization.

An active constraint can prevent the optimizer from moving in a direction that would otherwise decrease the objective.

1.6 A Simple Example Showing the Difference

Consider

$$\min_x x^2$$

subject to

$$x \geq 1.$$

The unconstrained minimum is

$$x = 0,$$

but $x = 0$ is infeasible.

The constrained minimum is

$$x^* = 1.$$

Notice that

$$f'(1) = 2 \neq 0.$$

Thus the unconstrained condition

$$f'(x^*) = 0$$

does not hold.

Why?

Because the direction toward $x = 0$ is forbidden.

The inequality constraint blocks the descent direction.

This is precisely the situation that KKT conditions describe.

1.7 Standard Form of a Constrained Optimization Problem

We will use the standard problem

$$\begin{array}{ll} \min_x & f(x) \\ \text{subject to} & g_i(x) \leq 0, \quad i = 1, \dots, m, \\ & h_j(x) = 0, \quad j = 1, \dots, p. \end{array}$$

The Lagrangian is

$$\mathcal{L}(x, \lambda, \mu) = f(x) + \sum_{i=1}^m \lambda_i g_i(x) + \sum_{j=1}^p \mu_j h_j(x).$$

For inequality constraints,

$$\lambda_i \geq 0.$$

For equality constraints,

$$\mu_j \in \mathbb{R}.$$

1.8 The Four KKT Conditions

Suppose x^* is an optimal solution and an appropriate constraint qualification holds. Then there exist multipliers

$$\lambda^* \in \mathbb{R}^m, \quad \mu^* \in \mathbb{R}^p$$

such that the following four conditions hold.

1.8.1 1. Stationarity

$$\nabla f(x^*) + \sum_{i=1}^m \lambda_i^* \nabla g_i(x^*) + \sum_{j=1}^p \mu_j^* \nabla h_j(x^*) = 0.$$

Equivalently,

$$\nabla_x \mathcal{L}(x^*, \lambda^*, \mu^*) = 0.$$

1.8.2 2. Primal Feasibility

The original constraints must be satisfied:

$$g_i(x^*) \leq 0, \quad i = 1, \dots, m,$$

and

$$h_j(x^*) = 0, \quad j = 1, \dots, p.$$

1.8.3 3. Dual Feasibility

For every inequality constraint,

$$\lambda_i^* \geq 0.$$

Equality multipliers remain unrestricted:

$$\mu_j^* \in \mathbb{R}.$$

1.8.4 4. Complementary Slackness

For every inequality constraint,

$$\lambda_i^* g_i(x^*) = 0.$$

These four groups of equations and inequalities are called the **Karush–Kuhn–Tucker conditions**.

1.9 Understanding Complementary Slackness

The condition

$$\lambda_i^* g_i(x^*) = 0$$

means that

$$\lambda_i^* = 0 \quad \text{or} \quad g_i(x^*) = 0.$$

Thus

$$g_i(x^*) < 0 \implies \lambda_i^* = 0.$$

Also,

$$\lambda_i^* > 0 \implies g_i(x^*) = 0.$$

Hence:

Important

Only active inequality constraints can have positive KKT multipliers.

The multiplier is therefore associated with a constraint that is actually binding at the optimum.

1.10 Geometric Interpretation of Stationarity

The stationarity condition is

$$\nabla f(x^*) + \sum_i \lambda_i^* \nabla g_i(x^*) + \sum_j \mu_j^* \nabla h_j(x^*) = 0.$$

Therefore

$$-\nabla f(x^*) = \sum_i \lambda_i^* \nabla g_i(x^*) + \sum_j \mu_j^* \nabla h_j(x^*).$$

The gradients of active constraints are normal to the corresponding constraint surfaces. Hence the negative objective gradient is a combination of constraint normals.

Intuition

At a constrained optimum, the objective wants to move in a descent direction, but the active constraints prevent that motion.

The constraint normals provide exactly the directions needed to balance the objective gradient.

In modern geometric language, this is related to the **normal cone** of the feasible set.

1.11 Why Must the Inequality Multipliers Be Nonnegative?

Suppose

$$g(x) \leq 0.$$

For a feasible point,

$$g(x) \leq 0.$$

If

$$\lambda \geq 0,$$

then

$$\lambda g(x) \leq 0.$$

Therefore

$$f(x) + \lambda g(x) \leq f(x).$$

Hence

$$\boxed{\mathcal{L}(x, \lambda) \leq f(x)}$$

for every feasible x .

This inequality will become the fundamental observation behind Lagrangian duality.

1.12 Numerical Example 1: One Inequality Constraint

Solve

$$\boxed{\min_x x^2 \quad \text{subject to} \quad x \geq 1.}$$

Rewrite the constraint as

$$1 - x \leq 0.$$

Thus

$$g(x) = 1 - x.$$

The Lagrangian is

$$\mathcal{L}(x, \lambda) = x^2 + \lambda(1 - x).$$

The KKT conditions are:

Stationarity

$$\frac{\partial \mathcal{L}}{\partial x} = 2x - \lambda = 0.$$

Thus

$$\lambda = 2x.$$

Primal Feasibility

$$1 - x \leq 0,$$

or

$$x \geq 1.$$

Dual Feasibility

$$\lambda \geq 0.$$

Complementary Slackness

$$\lambda(1 - x) = 0.$$

If

$$\lambda = 0,$$

then stationarity gives

$$x = 0,$$

which violates $x \geq 1$.

Therefore

$$1 - x = 0,$$

so

$$x^* = 1.$$

Then

$$\lambda^* = 2.$$

Hence

$$\boxed{x^* = 1, \quad \lambda^* = 2.}$$

The optimal value is

$$f(x^*) = 1.$$

1.13 Numerical Example 2: Two Variables and One Inequality

Solve

$$\boxed{\min_{x,y} [(x-2)^2 + (y-3)^2]}$$

subject to

$$x + y \geq 6.$$

Write the constraint as

$$g(x, y) = 6 - x - y \leq 0.$$

The Lagrangian is

$$\mathcal{L} = (x-2)^2 + (y-3)^2 + \lambda(6 - x - y).$$

Stationarity gives

$$2(x-2) - \lambda = 0,$$

and

$$2(y-3) - \lambda = 0.$$

Thus

$$x = 2 + \frac{\lambda}{2},$$

and

$$y = 3 + \frac{\lambda}{2}.$$

The unconstrained minimizer is

$$(2, 3),$$

but

$$2 + 3 = 5 < 6.$$

Therefore the constraint must be active:

$$x + y = 6.$$

Substitution gives

$$2 + \frac{\lambda}{2} + 3 + \frac{\lambda}{2} = 6.$$

Hence

$$5 + \lambda = 6,$$

so

$$\lambda = 1.$$

Therefore

$$x^* = \frac{5}{2}, \quad y^* = \frac{7}{2}.$$

Thus

$$\boxed{x^* = \frac{5}{2}, \quad y^* = \frac{7}{2}, \quad \lambda^* = 1.}$$

The optimal value is

$$f(x^*, y^*) = \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 = \frac{1}{2}.$$

1.14 Numerical Example 3: Two Inequality Constraints

Solve

$$\boxed{\min_{x,y} [(x-1)^2 + (y-2)^2]}$$

subject to

$$x + y \geq 4, \quad x \geq 0.$$

Write

$$g_1(x, y) = 4 - x - y \leq 0,$$

and

$$g_2(x, y) = -x \leq 0.$$

The Lagrangian is

$$\mathcal{L} = (x-1)^2 + (y-2)^2 + \lambda_1(4-x-y) - \lambda_2 x.$$

Stationarity:

$$2(x-1) - \lambda_1 - \lambda_2 = 0,$$

$$2(y-2) - \lambda_1 = 0.$$

Primal feasibility:

$$4 - x - y \leq 0, \quad -x \leq 0.$$

Dual feasibility:

$$\lambda_1, \lambda_2 \geq 0.$$

Complementary slackness:

$$\lambda_1(4 - x - y) = 0,$$

$$\lambda_2 x = 0.$$

The unconstrained minimizer is

$$(1, 2),$$

but

$$1 + 2 = 3 < 4.$$

Therefore the first constraint is active:

$$x + y = 4.$$

At the optimum, $x > 0$, so the second constraint is inactive:

$$\lambda_2 = 0.$$

Stationarity becomes

$$2(x - 1) = \lambda_1,$$

$$2(y - 2) = \lambda_1.$$

Hence

$$x - 1 = y - 2,$$

so

$$y = x + 1.$$

Together with

$$x + y = 4,$$

we obtain

$$x = \frac{3}{2}, \quad y = \frac{5}{2}.$$

Then

$$\lambda_1 = 1, \quad \lambda_2 = 0.$$

Therefore

$$\boxed{x^* = \frac{3}{2}, \quad y^* = \frac{5}{2}, \quad \lambda_1^* = 1, \quad \lambda_2^* = 0.}$$

This example illustrates complementary slackness particularly clearly.

1.15 General Interpretation of the KKT Multipliers

The multipliers associated with inequality constraints can be interpreted as measuring how strongly the corresponding constraints influence the optimum.

If

$$\lambda_i^* = 0,$$

the corresponding inequality is locally nonbinding.

If

$$\lambda_i^* > 0,$$

the constraint is active and influences the optimum.

This interpretation becomes particularly important in sensitivity analysis, duality, economics, machine learning, and mathematical programming.

1.16 Are KKT Conditions Always Necessary?

No.

A common misconception is

$$\text{optimal point} \implies \text{KKT conditions.}$$

This implication is not valid without additional regularity assumptions.

A suitable **constraint qualification** is generally required.

1.17 Constraint Qualifications

Constraint qualifications are assumptions that prevent pathological constraint geometries.

Important examples include:

- Linear Independence Constraint Qualification (LICQ),
- Mangasarian–Fromovitz Constraint Qualification (MFCQ),
- Constant Rank Constraint Qualification (CRCQ),
- Slater’s condition.

For a first course, LICQ and Slater’s condition are especially important.

1.18 Linear Independence Constraint Qualification (LICQ)

Let

$$A(x^*) = \{i : g_i(x^*) = 0\}$$

be the active inequality constraint set.

LICQ requires the vectors

$$\{\nabla g_i(x^*) : i \in A(x^*)\}$$

together with

$$\{\nabla h_j(x^*) : j = 1, \dots, p\}$$

to be linearly independent.

In other words, there should be no redundant active constraint normal.

Under standard smoothness assumptions, if x^* is a local optimum and LICQ holds, then KKT multipliers exist.

1.19 Counterexample: KKT Is Not Sufficient Even When LICQ Holds

Consider

$$\boxed{\min_x -x^2 \quad \text{subject to} \quad x^2 - 1 \leq 0.}$$

The feasible set is

$$[-1, 1].$$

We will show that $x^* = 0$ satisfies the KKT conditions and that LICQ holds, but $x^* = 0$ is not a minimum.

The Lagrangian is

$$\mathcal{L}(x, \lambda) = -x^2 + \lambda(x^2 - 1), \quad \lambda \geq 0.$$

Stationarity requires

$$\frac{\partial \mathcal{L}}{\partial x} = -2x + 2\lambda x = 2x(\lambda - 1) = 0.$$

At

$$x^* = 0,$$

choose

$$\lambda^* = 0.$$

Then all four KKT conditions hold:

- **Stationarity:**

$$-2(0) + 2(0)(0) = 0.$$

- **Primal feasibility:**

$$(0)^2 - 1 = -1 < 0.$$

- **Dual feasibility:**

$$\lambda^* = 0 \geq 0.$$

- **Complementary slackness:**

$$\lambda^* g(x^*) = 0(-1) = 0.$$

Thus

$$(x^*, \lambda^*) = (0, 0)$$

is a KKT point.

Now consider the constraint qualification. Since

$$g(0) = -1 < 0,$$

the constraint is inactive at $x^* = 0$. Therefore there are no active inequality constraints, and LICQ holds vacuously.

However,

$$f(0) = 0,$$

while, for example,

$$f(1) = -1.$$

Thus $x^* = 0$ is actually a local maximum, not a local minimum. The minima occur at

$$x = \pm 1.$$

Hence

KKT conditions are not sufficient for optimality, even when LICQ holds.

This example shows that a constraint qualification guarantees the *necessity* of KKT conditions at a local optimum, but does not make KKT conditions sufficient. Additional assumptions, such as convexity, are needed for sufficiency.

2 Lecture 2: Lagrangian Duality and Slater's Condition

2.1 From KKT to Duality

Consider the constrained problem

$$\begin{array}{ll} \min_x & f(x) \\ \text{subject to} & g_i(x) \leq 0, \quad i = 1, \dots, m, \\ & h_j(x) = 0, \quad j = 1, \dots, p. \end{array}$$

The Lagrangian is

$$\mathcal{L}(x, \lambda, \mu) = f(x) + \sum_i \lambda_i g_i(x) + \sum_j \mu_j h_j(x),$$

where

$$\lambda_i \geq 0.$$

Instead of fixing x and minimizing over the feasible set, we now fix the multipliers and minimize the Lagrangian over all x .

This leads to the **dual function**.

2.2 The Dual Function

Definition 2.1. *The Lagrangian dual function is*

$$q(\lambda, \mu) = \inf_x \mathcal{L}(x, \lambda, \mu),$$

where $\lambda \geq 0$.

Notice that the infimum is taken over *all* x , not just feasible x .

This is an important conceptual point.

The constraints have been incorporated into the Lagrangian rather than imposed directly on the minimization over x .

2.3 Why the Dual Function Gives a Lower Bound

Let x be any primal-feasible point.

Then

$$g_i(x) \leq 0$$

and

$$h_j(x) = 0.$$

Since

$$\lambda_i \geq 0,$$

we have

$$\lambda_i g_i(x) \leq 0.$$

Also,

$$\mu_j h_j(x) = 0.$$

Therefore

$$\mathcal{L}(x, \lambda, \mu) \leq f(x).$$

Since

$$q(\lambda, \mu) = \inf_x \mathcal{L}(x, \lambda, \mu),$$

we have

$$q(\lambda, \mu) \leq \mathcal{L}(x, \lambda, \mu) \leq f(x).$$

Thus

$$\boxed{q(\lambda, \mu) \leq f(x)}$$

for every feasible x .

If p^* denotes the primal optimal value, then

$$\boxed{q(\lambda, \mu) \leq p^*}.$$

This observation leads to weak duality.

2.4 The Dual Problem

The dual problem is defined as

$$\begin{array}{ll} \max_{\lambda, \mu} & q(\lambda, \mu) \\ \text{subject to} & \lambda \geq 0. \end{array}$$

The primal problem is a minimization problem while the dual problem is a maximization problem.

Let

$$p^*$$

denote the optimal value of the primal problem and

$$d^*$$

denote the optimal value of the dual problem.

2.5 Weak Duality Theorem

Theorem 2.2 (Weak Duality). *For any primal feasible point and any dual feasible multipliers,*

$$q(\lambda, \mu) \leq f(x).$$

Consequently,

$$\boxed{d^* \leq p^*}.$$

Proof. For any primal-feasible x ,

$$g_i(x) \leq 0, \quad h_j(x) = 0.$$

For any dual-feasible $\lambda \geq 0$,

$$\lambda_i g_i(x) \leq 0.$$

Hence

$$\mathcal{L}(x, \lambda, \mu) = f(x) + \sum_i \lambda_i g_i(x) + \sum_j \mu_j h_j(x) \leq f(x).$$

By definition,

$$q(\lambda, \mu) = \inf_x \mathcal{L}(x, \lambda, \mu) \leq \mathcal{L}(x, \lambda, \mu).$$

Therefore

$$q(\lambda, \mu) \leq f(x).$$

Since this holds for every feasible x and every dual-feasible (λ, μ) ,

$$d^* \leq p^*.$$

□

2.6 The Duality Gap

Define the duality gap as

$$p^* - d^*.$$

Weak duality guarantees

$$p^* - d^* \geq 0.$$

If

$$p^* - d^* = 0,$$

then the primal and dual optimal values coincide.

This is called **zero duality gap**.

2.7 Strong Duality

Definition 2.3. *Strong duality holds if*

$$p^* = d^*.$$

Thus:

$$\text{Weak duality} \quad : \quad d^* \leq p^*,$$

$$\text{Strong duality} \quad : \quad d^* = p^*.$$

Weak duality is essentially automatic from the definition of the dual.

Strong duality requires additional assumptions.

2.8 Numerical Example 4: Deriving a Dual Problem

Consider

$$\min_x \frac{1}{2}x^2 \quad \text{subject to} \quad x \geq 1.$$

Write the constraint as

$$1 - x \leq 0.$$

The Lagrangian is

$$\mathcal{L}(x, \lambda) = \frac{1}{2}x^2 + \lambda(1 - x), \quad \lambda \geq 0.$$

The dual function is

$$q(\lambda) = \inf_x \left[\frac{1}{2}x^2 + \lambda(1 - x) \right].$$

For fixed λ , minimize with respect to x :

$$\frac{\partial \mathcal{L}}{\partial x} = x - \lambda = 0.$$

Therefore

$$x = \lambda.$$

Substituting into the Lagrangian gives

$$q(\lambda) = \frac{1}{2}\lambda^2 + \lambda(1 - \lambda).$$

Hence

$$q(\lambda) = \lambda - \frac{1}{2}\lambda^2.$$

The dual problem is

$$\max_{\lambda \geq 0} \left(\lambda - \frac{1}{2} \lambda^2 \right).$$

Differentiate:

$$q'(\lambda) = 1 - \lambda.$$

Thus

$$\lambda^* = 1.$$

The dual optimal value is

$$d^* = 1 - \frac{1}{2} = \frac{1}{2}.$$

The primal optimum is

$$x^* = 1,$$

with

$$p^* = \frac{1}{2}.$$

Therefore

$$p^* = d^* = \frac{1}{2}.$$

Strong duality holds.

2.9 Slater's Condition

Consider the convex optimization problem

$$\begin{array}{ll} \min_x & f(x) \\ \text{subject to} & g_i(x) \leq 0, \quad i = 1, \dots, m, \\ & Ax = b. \end{array}$$

Assume:

- f is convex;
- every g_i is convex;
- equality constraints are affine.

Definition 2.4 (Slater's Condition). *Slater's condition holds if there exists a point $\tilde{x}$ such that*

$$g_i(\tilde{x}) < 0 \quad \text{for all } i$$

and

$$A\tilde{x} = b.$$

*Such a point is called a **strictly feasible point**.*

The strict inequality is essential:

$$g_i(\tilde{x}) < 0,$$

not merely

$$g_i(\tilde{x}) \leq 0.$$

2.10 Geometric Intuition for Slater's Condition

Slater's condition says that the feasible region contains a point that is strictly inside every inequality constraint while satisfying all equality constraints exactly.

In informal language:

Intuition

Slater's condition says that the convex feasible set has enough "interior room" and is not trapped entirely on the boundary of one or more inequality constraints.

It rules out many pathological constraint configurations.

2.11 Example: Slater's Condition Holds

Consider

$$\min_x x^2$$

subject to

$$x \geq 1.$$

Write

$$1 - x \leq 0.$$

Choose

$$\tilde{x} = 2.$$

Then

$$1 - 2 = -1 < 0.$$

Therefore Slater's condition holds.

The problem is convex, because

$$f(x) = x^2$$

is convex and

$$g(x) = 1 - x$$

is affine.

Hence the standard convex duality theorem applies.

2.12 Example: Slater's Condition Fails

Consider

$$\min_x x$$

subject to

$$x^2 \leq 0.$$

To satisfy Slater's condition we would need

$$x^2 < 0,$$

which is impossible.

Therefore Slater's condition fails.

Recall that this problem also provides an example where KKT fails at the optimal point.

This illustrates why regularity assumptions matter.

2.13 Strong Duality Under Slater's Condition

A fundamental result of convex optimization is the following.

Theorem 2.5 (Slater's Strong Duality Theorem). *Consider*

$$\begin{aligned} \min_x \quad & f(x) \\ \text{subject to} \quad & g_i(x) \leq 0, \quad i = 1, \dots, m, \\ & Ax = b, \end{aligned}$$

where f and g_i are convex and the equality constraints are affine.

If Slater's condition holds, then, under the standard finite-valued feasibility assumptions,

$$\boxed{p^* = d^*}.$$

Thus strong duality holds.

This theorem is one of the central results connecting convex optimization, duality, and KKT conditions.

2.14 Why Convexity Matters

For a general nonconvex problem, satisfying KKT conditions does not necessarily mean that the point is a global minimum.

For example, a KKT point may be:

- a local minimum,
- a local maximum,
- a saddle point,
- or another stationary-type point.

Convexity changes the situation dramatically.

For a convex optimization problem, every local minimum is a global minimum.

Moreover, under the appropriate assumptions, KKT conditions become sufficient for global optimality.

2.15 KKT Sufficiency for Convex Problems

Consider

$$\begin{aligned} \min_x \quad & f(x) \\ \text{subject to} \quad & g_i(x) \leq 0, \quad i = 1, \dots, m, \\ & Ax = b. \end{aligned}$$

Assume:

- f is convex and differentiable;
- each g_i is convex and differentiable;
- $Ax = b$ is affine.

Suppose x^* and multipliers $\lambda^* \geq 0$ satisfy KKT.

We will show that x^* is a global optimum.

2.16 Proof of KKT Sufficiency

Convexity of f gives

$$f(x) \geq f(x^*) + \nabla f(x^*)^\top (x - x^*).$$

From stationarity,

$$\nabla f(x^*) = - \sum_i \lambda_i^* \nabla g_i(x^*) - A^\top \mu^*.$$

Therefore

$$\begin{aligned} f(x) &\geq f(x^*) - \sum_i \lambda_i^* \nabla g_i(x^*)^\top (x - x^*) \\ &\quad - \mu^{*\top} A(x - x^*). \end{aligned}$$

Because both x and x^* satisfy the affine equality constraints,

$$Ax = b, \quad Ax^* = b.$$

Hence

$$A(x - x^*) = 0.$$

Therefore

$$\mu^{*\top} A(x - x^*) = 0.$$

Thus

$$f(x) \geq f(x^*) - \sum_i \lambda_i^* \nabla g_i(x^*)^\top (x - x^*).$$

Since g_i is convex,

$$g_i(x) \geq g_i(x^*) + \nabla g_i(x^*)^\top (x - x^*).$$

Hence

$$\nabla g_i(x^*)^\top (x - x^*) \leq g_i(x) - g_i(x^*).$$

Multiplying by $\lambda_i^* \geq 0$ gives

$$\lambda_i^* \nabla g_i(x^*)^\top (x - x^*) \leq \lambda_i^* [g_i(x) - g_i(x^*)].$$

Since x is feasible,

$$g_i(x) \leq 0.$$

Complementary slackness gives

$$\lambda_i^* g_i(x^*) = 0.$$

Therefore

$$\lambda_i^* [g_i(x) - g_i(x^*)] \leq 0.$$

Hence

$$\lambda_i^* \nabla g_i(x^*)^\top (x - x^*) \leq 0.$$

Consequently,

$$f(x) \geq f(x^*).$$

Since this holds for every feasible x ,

$x^* \text{ is a global optimum.}$

Thus KKT conditions are sufficient for global optimality in this convex setting.

2.17 The Fundamental Convex KKT Theorem

Combining necessity and sufficiency gives the central result.

Theorem 2.6 (Convex KKT Characterization). *Consider*

$$\begin{aligned} \min_x \quad & f(x) \\ \text{subject to} \quad & g_i(x) \leq 0, \quad i = 1, \dots, m, \\ & Ax = b. \end{aligned}$$

Suppose f and g_i are convex and differentiable and the equality constraints are affine. If an appropriate constraint qualification such as Slater's condition holds, then

$$\boxed{x^* \text{ is optimal} \iff \exists (\lambda^*, \mu^*) \text{ satisfying KKT.}}$$

In particular, Slater's condition gives strong duality:

$$\boxed{p^* = d^*}.$$

This is one of the most important facts in convex optimization.

2.18 Relationship Between KKT and Duality

Suppose

$$(x^*, \lambda^*, \mu^*)$$

satisfies KKT.

Stationarity gives

$$\nabla_x \mathcal{L}(x^*, \lambda^*, \mu^*) = 0.$$

For a convex problem, $\mathcal{L}$ is convex in x because it is a nonnegative linear combination of convex inequality functions plus affine equality functions and the convex objective.

Therefore x^* minimizes the Lagrangian:

$$q(\lambda^*, \mu^*) = \mathcal{L}(x^*, \lambda^*, \mu^*).$$

Now use complementary slackness and primal feasibility:

$$\lambda_i^* g_i(x^*) = 0,$$

and

$$h_j(x^*) = 0.$$

Therefore

$$\begin{aligned} \mathcal{L}(x^*, \lambda^*, \mu^*) &= f(x^*) + \sum_i \lambda_i^* g_i(x^*) + \sum_j \mu_j^* h_j(x^*) \\ &= f(x^*). \end{aligned}$$

Thus

$$q(\lambda^*, \mu^*) = f(x^*).$$

If x^* is primal optimal,

$$f(x^*) = p^*.$$

Hence

$$q(\lambda^*, \mu^*) = p^*.$$

Since weak duality gives

$$d^* \leq p^*,$$

we obtain

$$\boxed{d^* = p^*}.$$

Thus the KKT conditions provide a direct bridge between primal and dual optimality.

2.19 Numerical Example 5: Quadratic Programming

Consider

$$\min_{x,y} \frac{1}{2}(x^2 + y^2)$$

subject to

$$x + y \geq 2.$$

Write

$$g(x, y) = 2 - x - y \leq 0.$$

The Lagrangian is

$$\mathcal{L} = \frac{1}{2}(x^2 + y^2) + \lambda(2 - x - y).$$

Stationarity gives

$$x - \lambda = 0,$$

$$y - \lambda = 0.$$

Hence

$$x = y = \lambda.$$

The unconstrained minimum is $(0, 0)$, which violates

$$x + y \geq 2.$$

Thus the constraint is active:

$$x + y = 2.$$

Therefore

$$2\lambda = 2,$$

so

$$\lambda = 1.$$

Hence

$$x^* = 1, \quad y^* = 1, \quad \lambda^* = 1.$$

The primal objective value is

$$p^* = \frac{1}{2}(1 + 1) = 1.$$

The dual function can be derived:

$$q(\lambda) = \inf_{x,y} \left[\frac{1}{2}(x^2 + y^2) + \lambda(2 - x - y) \right].$$

Stationarity with respect to x, y gives

$$x = y = \lambda.$$

Substitution yields

$$q(\lambda) = \lambda - \lambda^2.$$

The dual problem is

$$\max_{\lambda \geq 0} (\lambda - \lambda^2).$$

The maximizer is

$$\lambda^* = \frac{1}{2}.$$

However, note that the exact numerical coefficient here depends on the scaling of the original quadratic and constraint representation. Therefore one must always derive the dual carefully from the exact Lagrangian rather than relying on a memorized formula.

For the original problem above, directly substituting $x = y = \lambda$ gives

$$\mathcal{L} = \lambda^2 + \lambda(2 - 2\lambda) = 2\lambda - \lambda^2.$$

Hence

$$q(\lambda) = 2\lambda - \lambda^2,$$

whose maximum occurs at

$$\lambda^* = 1.$$

Therefore

$$d^* = 1 = p^*.$$

This example illustrates the importance of checking the algebra in the dual derivation.

2.20 Shadow Price Interpretation of Multipliers

Suppose a constraint depends on a parameter b :

$$g(x) \leq b.$$

Let the optimal value be

$$p^*(b).$$

Under suitable regularity conditions, the optimal multiplier measures the marginal sensitivity of the optimal value with respect to the constraint parameter.

The exact sign depends on how the constraint is written.

For example, if we write

$$g(x) - b \leq 0,$$

then the Lagrangian contains

$$\lambda[g(x) - b].$$

Under appropriate differentiability conditions,

$$\frac{dp^*}{db}$$

is related to the optimal multiplier, with the sign determined by this representation.

The multiplier is often called a **shadow price**.

Intuition

A large multiplier means that changing the corresponding constraint can have a significant effect on the optimal objective value.

A zero multiplier indicates that, locally, relaxing that constraint does not change the optimal value to first order.

3 Applications of KKT and Duality in Machine Learning

3.1 Why KKT Matters in Machine Learning

Many machine learning problems can be written as optimization problems.

For example,

$$\min_{\theta} L(\theta)$$

is an unconstrained optimization problem.

But many important models involve constraints:

$$\min_{\theta} L(\theta) \quad \text{subject to constraints on } \theta.$$

KKT conditions provide a systematic way to characterize optimal solutions to such constrained problems.

Duality is equally important because many ML problems have dual formulations that are computationally or theoretically more convenient.

3.2 Support Vector Machines

The hard-margin Support Vector Machine can be formulated as

$$\min_{w,b} \frac{1}{2} \|w\|^2$$

subject to

$$y_i(w^\top x_i + b) \geq 1, \quad i = 1, \dots, n.$$

Rewrite the constraints as

$$1 - y_i(w^\top x_i + b) \leq 0.$$

Introduce multipliers

$$\alpha_i \geq 0.$$

The Lagrangian is

$$\mathcal{L}(w, b, \alpha) = \frac{1}{2} \|w\|^2 + \sum_{i=1}^n \alpha_i [1 - y_i(w^\top x_i + b)].$$

The KKT complementary slackness condition is

$$\alpha_i [1 - y_i(w^\top x_i + b)] = 0.$$

Therefore:

If

$$\alpha_i > 0,$$

then

$$y_i(w^\top x_i + b) = 1.$$

Such points are the **support vectors**.

If

$$y_i(w^\top x_i + b) > 1,$$

then

$$\alpha_i = 0.$$

Thus complementary slackness directly explains the role of support vectors.

$$\boxed{\text{KKT} \longrightarrow \text{complementary slackness} \longrightarrow \text{support vectors.}}$$

3.3 LASSO and KKT

The LASSO problem is

$$\min_{\beta} \frac{1}{2} \|X\beta - y\|^2 + \lambda \|\beta\|_1.$$

The ℓ_1 term promotes sparsity.

Because the ℓ_1 norm is not differentiable at zero, subgradient conditions are used.

For $\beta_j \neq 0$,

$$\partial|\beta_j| = \text{sign}(\beta_j).$$

At $\beta_j = 0$,

$$\partial|\beta_j| = [-1, 1].$$

Therefore the optimality conditions can be expressed through subgradients.

This explains why coefficients can become exactly zero.

Thus KKT-type optimality ideas are fundamental to sparse optimization.

3.4 Quadratic Programming

Many machine learning problems reduce to quadratic programs:

$$\min_x \frac{1}{2} x^T Q x + c^T x$$

subject to

$$Ax = b, \quad Cx \leq d.$$

The KKT conditions form a system involving the primal variables and Lagrange multipliers.

For equality-constrained quadratic programming,

$$\min_x \frac{1}{2} x^T A x - b^T x$$

subject to

$$Cx = d,$$

the KKT system is

$$\begin{bmatrix} A & C^T \\ C & 0 \end{bmatrix} \begin{bmatrix} x \\ \mu \end{bmatrix} = \begin{bmatrix} b \\ d \end{bmatrix}.$$

Large versions of such systems can be solved using iterative numerical methods.

3.5 KKT and Conjugate Gradient

KKT and Conjugate Gradient are not competing ideas.

They answer different questions.

Important

KKT: What conditions must an optimum satisfy?

Conjugate Gradient: How can certain large linear systems be solved efficiently?

A constrained quadratic optimization problem may first be converted using KKT into a linear system, and then an appropriate iterative solver may be used.

Thus the conceptual chain can be

Constrained optimization $\rightarrow$ KKT system $\rightarrow$ large linear system $\rightarrow$ iterative solver.

4 Practical Procedure for Solving KKT Problems

Consider

$$\begin{aligned} \min_x \quad & f(x) \\ \text{subject to} \quad & g_i(x) \leq 0, \quad i = 1, \dots, m, \\ & h_j(x) = 0, \quad j = 1, \dots, p. \end{aligned}$$

Use the following procedure.

Step 1: Put Every Inequality in Standard Form

Write

$$g_i(x) \leq 0.$$

For example,

$$x \geq 1$$

becomes

$$1 - x \leq 0.$$

This step is important because the sign of the multiplier depends on the chosen convention.

Step 2: Construct the Lagrangian

Write

$$\mathcal{L} = f + \sum_i \lambda_i g_i + \sum_j \mu_j h_j.$$

Step 3: Write Stationarity

Calculate

$$\nabla_x \mathcal{L} = 0.$$

Step 4: Write Primal Feasibility

$$g_i(x) \leq 0, \quad h_j(x) = 0.$$

Step 5: Write Dual Feasibility

$$\lambda_i \geq 0.$$

Step 6: Write Complementary Slackness

$$\lambda_i g_i(x) = 0.$$

Step 7: Identify Active and Inactive Constraints

For every inequality, determine whether

$$g_i(x) = 0$$

or

$$g_i(x) < 0.$$

If

$$g_i(x) < 0,$$

then

$$\lambda_i = 0.$$

Step 8: Solve the Resulting Equations

Solve the stationarity, active constraints, and multiplier equations.

Step 9: Check Every KKT Condition

A candidate is not a solution merely because stationarity holds.

Verify:

Stationarity:	$\nabla_x \mathcal{L} = 0,$
Primal feasibility:	$g_i(x) \leq 0, \quad h_j(x) = 0,$
Dual feasibility:	$\lambda_i \geq 0,$
Complementary slackness:	$\lambda_i g_i(x) = 0.$

5 Important Distinctions Students Must Remember**5.1 KKT Is Not Automatically Necessary**

For a general nonlinear optimization problem,

$$x^* \text{ optimal}$$

does not automatically imply KKT.

A constraint qualification is needed.

For example, LICQ can provide the required regularity.

5.2 KKT Is Not Automatically Sufficient

For a nonconvex problem, satisfying KKT does not necessarily imply global optimality.

Convexity is crucial.

5.3 Convexity + Appropriate Constraint Qualification

For a convex problem satisfying the appropriate regularity assumptions, KKT becomes a powerful characterization of global optimality.

In particular, under the standard convex assumptions and Slater's condition,

$x^* \text{ optimal} \iff x^* \text{ satisfies KKT.}$

5.4 Slater Is Not the Same as KKT

Slater's condition is a **constraint qualification**.

KKT conditions are **optimality conditions**.

Slater helps establish strong duality and, in the convex setting, helps ensure the existence of KKT multipliers.

6 Theorem Sheet

6.1 General KKT Conditions

For

$$\begin{aligned} \min_x \quad & f(x) \\ \text{subject to} \quad & g_i(x) \leq 0, \\ & h_j(x) = 0, \end{aligned}$$

define

$$\mathcal{L}(x, \lambda, \mu) = f(x) + \sum_i \lambda_i g_i(x) + \sum_j \mu_j h_j(x).$$

KKT conditions are

$$\nabla_x \mathcal{L}(x^*, \lambda^*, \mu^*) = 0$$

$$g_i(x^*) \leq 0, \quad h_j(x^*) = 0$$

$$\lambda_i^* \geq 0$$

$$\lambda_i^* g_i(x^*) = 0.$$

6.2 Weak Duality

$$d^* \leq p^*.$$

This holds for the Lagrangian dual problem under the standard formulation.

6.3 Strong Duality

$$d^* = p^*.$$

6.4 Duality Gap

$$p^* - d^* \geq 0.$$

6.5 Slater's Condition

For a convex problem

$$\begin{aligned} \min_x \quad & f(x) \\ \text{subject to} \quad & g_i(x) \leq 0, \\ & Ax = b, \end{aligned}$$

Slater requires some $\tilde{x}$ such that

$$g_i(\tilde{x}) < 0 \quad \forall i, \quad A\tilde{x} = b.$$

6.6 Convex KKT Theorem

Under the standard differentiability, convexity, feasibility, and regularity assumptions,

$$x^* \text{ is globally optimal} \iff \exists(\lambda^*, \mu^*) \text{ satisfying KKT.}$$

6.7 Complementary Slackness

$$\lambda_i g_i(x) = 0.$$

Therefore

$$g_i(x) < 0 \Rightarrow \lambda_i = 0.$$

And

$$\lambda_i > 0 \Rightarrow g_i(x) = 0.$$

7 The Big Picture

The development of constrained optimization can be summarized as follows.

$$\boxed{\text{Unconstrained optimization}}$$

$$\nabla f(x^*) = 0$$

$\Downarrow$

$$\boxed{\text{Equality constraints}}$$

$$\nabla f(x^*) + \sum_j \mu_j \nabla h_j(x^*) = 0$$

$\Downarrow$

$$\boxed{\text{Inequality constraints}}$$

$\Downarrow$

$$\boxed{\text{Active and inactive constraints}}$$

$\Downarrow$

$$\boxed{\text{KKT conditions}}$$

$\Downarrow$

$$\boxed{\text{Lagrangian duality}}$$

$\Downarrow$

$$\boxed{d^* \leq p^*}$$

$\Downarrow$

Slater's condition

 $\Downarrow$

$d^* = p^*$

 $\Downarrow$

KKT characterizes global optimality for convex problems.

8 Final Intuitive Interpretation

The entire subject can be understood through a simple question:

Where is the lowest point that I am actually allowed to reach?

For an unconstrained problem, the answer is characterized by

$$\nabla f(x^*) = 0.$$

For a constrained problem, some directions are forbidden.

At an optimum, there can therefore be a nonzero objective gradient, provided that the descent direction is blocked by active constraints.

The KKT stationarity condition says

$\nabla f + \text{weighted constraint normals} = 0.$

The multipliers are the weights.

Complementary slackness says

$\text{inactive constraint} \implies \text{zero multiplier}.$

Convexity gives the crucial global interpretation:

$\text{KKT point} \implies \text{global optimum}.$

Slater's condition provides a regularity condition ensuring clean duality behavior.

Duality then connects the primal minimization problem with a dual maximization problem:

$\text{primal minimum} \geq \text{dual maximum}.$

Under strong duality,

$\text{primal minimum} = \text{dual maximum}.$

Thus the conceptual chain is

$\text{Geometry} \rightarrow \text{Lagrange multipliers} \rightarrow \text{KKT} \rightarrow \text{Duality} \rightarrow \text{Slater} \rightarrow \text{Convex optimality}.$

9 Suggested Two-Lecture Teaching Plan

9.1 Lecture 1: KKT Conditions

A possible 90-minute structure is:

Time	Topic
10 min	Why $\nabla f = 0$ fails in constrained optimization
15 min	Equality constraints and geometric Lagrange multipliers
15 min	Inequality constraints
10 min	Active and inactive constraints
15 min	Derivation and interpretation of KKT
10 min	Complementary slackness
10 min	Numerical examples
5 min	Summary and questions

The most important transition is

$$\text{interior optimum} \Rightarrow \nabla f = 0$$

versus

$$\text{boundary optimum} \Rightarrow -\nabla f \text{ is balanced by active constraint normals.}$$

9.2 Lecture 2: Duality and Slater

A possible 90-minute structure is:

Time	Topic
10 min	Recap of KKT
20 min	Lagrangian and dual function
15 min	Weak duality and proof
15 min	Strong duality
15 min	Slater's condition
10 min	KKT sufficiency for convex problems
5 min	SVM and LASSO connections

10 One-Page Quick Reference

Primal Problem

$$\begin{array}{ll} \min_x & f(x) \\ \text{s.t.} & g_i(x) \leq 0, \\ & h_j(x) = 0. \end{array}$$

Lagrangian

$$\mathcal{L}(x, \lambda, \mu) = f(x) + \sum_i \lambda_i g_i(x) + \sum_j \mu_j h_j(x).$$

KKT

$$\nabla_x \mathcal{L} = 0$$

$$g_i(x) \leq 0, \quad h_j(x) = 0$$

$$\lambda_i \geq 0$$

$$\lambda_i g_i(x) = 0.$$

Dual Function

$$q(\lambda, \mu) = \inf_x \mathcal{L}(x, \lambda, \mu).$$

Dual Problem

$$\max_{\lambda \geq 0, \mu} q(\lambda, \mu).$$

Weak Duality

$$d^* \leq p^*.$$

Strong Duality

$$d^* = p^*.$$

Slater

$$\exists \tilde{x} : g_i(\tilde{x}) < 0, \quad A\tilde{x} = b.$$

Convex KKT

Under the appropriate convexity and regularity assumptions,

$$x^* \text{ optimal} \iff \exists (\lambda^*, \mu^*) \text{ satisfying KKT.}$$

11 Memory Aid for Students

The four KKT conditions can be remembered as

$$\boxed{\text{S-P-D-C}}$$

1. S – Stationarity

$$\nabla_x \mathcal{L} = 0.$$

2. P – Primal feasibility

$$g_i(x) \leq 0, \quad h_j(x) = 0.$$

3. **D – Dual feasibility**

$$\lambda_i \geq 0.$$

4. **C – Complementary slackness**

$$\lambda_i g_i(x) = 0.$$

A useful conceptual memory aid is:

Stationarity $\Downarrow$ No feasible descent direction

Primal feasibility $\Downarrow$ The point is allowed
--

Dual feasibility $\Downarrow$ Correct multiplier signs
--

Complementary slackness $\Downarrow$ Only binding constraints matter
--

Finally,

$\text{KKT} + \text{Convexity} + \text{appropriate regularity} \implies \text{Global optimality.}$
--